

Problem Statement ID – 26035

Problem Statement Title - Development of a Software Program/Application for Generation of Test Reports for Non-Automatic Weighing Instruments (NAWI) as per OIML Recommendation R- 76

Description:

- **Background:**

Non-Automatic Weighing Instruments (NAWIs), such as electronic weighing scales, platform scales and weighbridges, are widely used in trade, commerce, healthcare, agriculture and industry where accurate measurement is essential for fair transactions and consumer protection. Under the Legal Metrology Act, 2009 and the Legal Metrology (General) Rules, 2011, such instruments used for transaction and protection are required to conform to prescribed standards, obtain model approval, and undergo verification and stamping before being put into use.

For granting model approval, NAWIs are evaluated by designated laboratories in accordance with OIML Recommendation R 76 “Non-Automatic Weighing Instruments, which specifies internationally accepted technical, metrological and performance requirements. The evaluation involves various metrological and functional tests, the results of which are compiled into detailed test reports. At present, these test reports are largely prepared manually using spreadsheets or document templates, making the process time-consuming, prone to calculation errors and lacking uniformity. Hence, there is a requirement to develop a software application that automates test data recording, compliance evaluation and generation of standardized test reports as per OIML R 76, thereby improving accuracy, consistency, and efficiency in the model approval process.

- **Description:**

Develop a software application capable of generating complete test reports for Non-Automatic Weighing Instruments based on test observations recorded during type evaluation as per OIML R 76.

The system should be capable of: - Capturing instrument details and technical specifications.

- Recording laboratory and environmental conditions.
- Entering observations from various OIML R 76 test procedures.
- Automatically calculating permissible errors, and compliance status.
- Performing validation checks for entered test data.
- Automatically determining pass/fail criteria based on OIML R 76 requirements.

- Generating standardized digital test reports in printable formats.
 - Maintaining a digital repository of completed test reports.
 - Providing secure user access with role-based permissions.
 - Supporting future updates whenever OIML recommendations are revised.
- **Expected Solution:**

The proposed solution should include:

- User-friendly desktop and/or web-based application.
- Digital data entry forms for all applicable OIML R 76 tests.
- Automated calculations and compliance verification.
- Standardized test report generation in PDF and editable formats MS Word etc.
- Instrument-wise test history and report repository.
- Dashboard for monitoring testing activities and report status.
- Search and retrieval facility for previously generated reports.
- Technical documentation describing software architecture, calculation methodology and deployment framework.

Key Functional Requirements:

- Entry of manufacturer details, Instrument specifications, Model information and technical parameters
- Compliance determination as per OIML R-76
- Entry of observations for all prescribed tests
- Automatic validation of input data and related calculations
- Automatic preparation of standardized test reports with auto-population of laboratory and instrument details
- Attachment of photographs and supporting documents
- Digital signatures (optional)
- Export to PDF and editable formats
- Dashboard for test report management (completed, in process, history access etc.)

Organization: Ministry of Consumer Affairs, Food & Public Distribution

Department: Department of Consumer Affairs (DoCA)

Category: Software

Dataset link: <https://consumeraffairs.gov.in/pages/legal-metrology-act>